

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

GRADUATION THESIS

LLM-based Emotion Recognition in Conversation

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Program: Data Science & Artificial Intelligence

Supervisor: Doctor of Philosophy Nguyen Kiem Hieu

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ABSTRACT

Emotion Recognition in Conversations (ERC) is a challenging task because human emotions are often mixed, subtle, or ambiguous. Traditional approaches suffer from two major limitations: they force models to output a single discrete label (hard labeling), discarding emotional ambiguity, and they rely on monolithic multimodal fusion, which often suffers from modality collapse and produces opaque decisions. This thesis proposes a soft-label, multi-agent framework that addresses both limitations. The framework decomposes the ERC task into three specialized unimodal agents (text, acoustic, visual), each outputting a probability distribution over emotion classes together with a natural language rationale. A central resolver agent then synthesizes these inputs through symbolic, text-based reasoning, explicitly weighing cross-modal consensus and resolving conflicts using modality-specific priors. All agents are powered by Gemini-2.5-flash-lite. Experiments on the MELD and IEMOCAP benchmarks show that soft labeling improves accuracy by 2-8% across all configurations, and the agentic resolver outperforms monolithic fusion by 5.65% absolute accuracy on MELD and 1.66% on IEMOCAP. Qualitative analysis reveals that the resolver's advantage stems from independent specialist agents and contained modality failures. These results demonstrate that soft labeling preserves emotional ambiguity, and that symbolic, text-based multi-agent reasoning is a viable alternative to dense vector fusion for multimodal ERC.

Student

(Signature and full name)

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LIST OF ABBREVIATIONS

Abbreviation	Definition
CoT	Chain-of-Thought
eGeMAPS	extended Geneva Minimalistic Acoustic Parameter Set
ERC	Emotion Recognition in Conversations
GRU	Gated Recurrent Unit
IEMOCAP	Interactive Emotional Dyadic Motion Capture
JSD	Jensen-Shannon Divergence
KL	Kullback-Leibler Divergence
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MAS	Multi-Agent System
MELD	Multimodal EmotionLines Dataset
MLLM	Multimodal Large Language Model

CHAPTER 1. INTRODUCTION

1.1 Problem Statement

Human communication is inherently multimodal, combining verbal language, vocal acoustics, and visual expressions. While conventional single-modality models have advanced significantly, they fail in natural conversations because they ignore critical cross-modal information. Speakers rarely speak their minds openly; instead they often use sarcasm, deception, emotion masking, or suppression. A model relying solely on text transcripts is therefore highly prone to misclassification.

To formalize this challenge, let a conversation D be a sequence of time-ordered utterances:

$$D = \{u_1, u_2, \dots, u_N\} \quad (1.1)$$

For a target utterance u_i , its historical context within a window of size w is:

$$C_i = \{u_{i-w}, u_{i-w+1}, \dots, u_{i-1}\} \quad (1.2)$$

Each utterance u_i is a multimodal tuple containing three parallel, time-synchronized streams:

$$u_i = (t_i, a_i, v_i) \quad (1.3)$$

where t_i is the textual transcript, a_i the raw acoustic waveform, and v_i the video frame sequence.

Traditional ERC approaches introduce two core bottlenecks.

First, categorical collapse (the hard-label problem). Most systems treat emotion classification as a mutually exclusive labeling task, where a model maps inputs to a discrete label:

$$f(u_i, C_i) \rightarrow y \quad (1.4)$$

where $y \in \mathcal{E}$ is a one-hot vector over K emotion classes. Forcing models to choose a single label discards valuable information about mixed, subtle, or ambiguous emotional states.

Second, modality blurring in monolithic fusion. Traditional architectures use a single large language model or multimodal LLM that processes text, audio, and

visual features together in one forward pass:

$$P = \mathcal{M}(t_i, a_i, v_i, C_i) \quad (1.5)$$

where $\mathcal{M}$ denotes the black-box LLM that encodes all modalities and produces a categorical distribution $P = [p_1, \dots, p_K]$ over the emotion space. This dense fusion creates a heavy cognitive load, leading to *modality dominance*, in which the model favors one modality and produces opaque decisions that fail when a single modality contains noise.

The core research problem is: *How can we design a multimodal emotion recognition framework that preserves emotional ambiguity and achieves explainable conflict resolution without informational collapse?*

1.2 Background and Problems of Research

The pursuit of human-centered artificial intelligence has placed emotional intelligence at the center of modern human-computer interaction. Emotion Recognition in Conversation (ERC) has consequently emerged as a vital research domain, enabling applications in digital mental health, empathetic virtual assistants, intelligent education, and call-center analytics. Unlike standard classification tasks, ERC requires tracking emotional trajectories over sequential speaker turns, making context and linguistic cues essential.

Despite its importance, the academic literature reveals two persistent problems that prevent multimodal ERC systems from achieving real-world reliability.

First, there is the problem of categorical collapse, or the hard-label problem. Most current ERC systems treat emotion classification as a hard, mutually exclusive labeling task. By forcing models to output one-hot vectors, they cannot represent mixed, subtle, or ambiguous human emotions. This loss of nuance leads to rigid and inaccurate emotional tracking.

Second, modality alignment limits arise from monolithic fusion architectures. Traditional multimodal models rely on monolithic MLLMs that attempt to process and merge text, audio, and visual features simultaneously. Training these models requires large amounts of paired data, and they frequently suffer from modality dominance - the model overly relies on signals from one specific modality while ignoring others.

1.2.1 Research Hypotheses and Core Contributions

To address these bottlenecks, this thesis proposes a shift in how multimodal conversational emotions are represented, fused, and processed. We structure our

investigation around two core research hypotheses.

Hypothesis 1 (the soft-label boost): Modeling conversational emotional states as continuous probability distributions (soft labels) rather than discrete categorical variables prevents premature information loss and yields a consistent performance improvement across all unimodal and multimodal configurations.

Hypothesis 2 (agentic vs. monolithic superiority): Decomposing the complex multimodal ERC task into independent, specialized component agents (text, audio, visual), coupled with a resolver agent that performs linguistic reasoning, outperforms dense, single-block monolithic fusion architectures under identical unsupervised conditions.

By systematically testing these hypotheses, this thesis delivers two primary contributions to the field of affective computing.

First, we provide empirical validation of the soft-label paradigm. We demonstrate that converting hard labels into continuous probability distributions leads to systematic improvements in accuracy and F1 score across all tested configurations (unimodal text, unimodal visual, unimodal acoustic, multimodal, and agentic settings) on the MELD benchmark.

Second, we establish an explainable, zero-shot multi-agent coordination framework. We propose and validate a modular architecture that replaces traditional black-box vector fusion with symbolic, text-based reasoning using frozen model backbones. By evaluating conversational inputs through three specialized unimodal agents whose outputs are integrated by a central resolver agent, our system isolates modality failures and provides a human-auditable decision trail. While operating completely out-of-the-box without target-domain training weights, this training-free framework successfully outperforms internal zero-shot monolithic baselines by up to 5.65% absolute accuracy and narrows the gap against supervised state-of-the-art models to within single digits on short-form transcripts.

1.3 Objectives and Scope

The primary objective of this research is to design, implement, and benchmark an explainable, robust, soft-label-driven agentic framework for multimodal conversational emotion recognition.

The empirical evaluations focus on two benchmark datasets. The first is MELD (Multimodal EmotionLines Dataset) [1], which contains short, multi-party dialogue segments taken from the *Friends* TV show. MELD is characterized by high background noise, frequent speaker overlaps, sarcasm, and open-domain conver-

sational topics, making it a challenging testing ground with a low signal-to-noise ratio. The second is IEMOCAP (Interactive Emotional Dyadic Motion Capture) [2], which contains long, dyadic (two-person) acted and spontaneous interactions. IEMOCAP requires deep context lookback over historical conversational turns to resolve emotional ambiguity.

The system uses the Gemini-2.5-flash-lite model [3] as the backbone due to its native text, video, and audio input capabilities. We evaluate baseline unimodal performance, monolithic fusion models, heuristic-based fusion, and our proposed soft-label agentic resolver. The evaluation tracks standard classification metrics (accuracy, weighted F1, top-2 accuracy) alongside distribution similarity metrics (Jensen-Shannon divergence).

1.4 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 provides a literature review of classical discrete and continuous emotion theories, critiques existing early and late monolithic fusion pipelines, explores the history of speech emotion recognition, and catalogs state-of-the-art large language model approaches in affective computing.

Chapter 3 presents the methodology, describing the complete technical design of our proposed soft-label multi-agent system. It details the specialized unimodal semantic (text), acoustic, and visual component agents, and the chain-of-thought (CoT) prompting structure of the resolver agent.

Chapter 4 reports the numerical results, forming the empirical core of this thesis. It presents side-by-side results showing the universal benefit of soft labeling, the superiority of agentic reasoning over monolithic architectures, and the improved performance of raw waveform audio LLMs over eGeMAPS. It also provides deep-dive analyses on visual dominance, agentic resilience, and data integrity constraints.

Chapter 5 concludes the thesis by summarizing the core contributions, highlighting practical system bottlenecks (latency and API costs), and outlining a path forward using knowledge distillation to port agentic soft-label benefits into compact local models.

CHAPTER 2. LITERATURE REVIEW

Chapter 1 introduced the research problem and outlined the limitations of existing ERC systems. This chapter provides a detailed review of the foundational literature. It first examines psychological theories of emotion, then surveys classical and state-of-the-art approaches to ERC, including hard versus soft labeling, monolithic multimodal fusion, and emerging agentic architectures.

2.1 Hard Labeling Versus Soft Labeling in Emotion Recognition

Early theories in cognitive psychology, such as Ekman’s six basic emotions (anger, disgust, fear, happiness, sadness, surprise) [4] and Plutchik’s emotion wheel [5], suggest that human emotions fall into distinct, universal categories. Following this view, natural language processing research has traditionally treated emotion recognition as a single-label classification problem. Models are trained to map an input to one mutually exclusive class. While this hard-labeling approach works pretty well, it has a major flaw: real emotions are rarely discrete or mutually exclusive.

Russell’s Circumplex Model of Affect [6] offers an alternative. It describes emotions as continuous states along two dimensions: valence (pleasantness) and arousal (intensity). In real conversations, people often feel mixed or subtle emotions (sarcastic amusement, bittersweetness, anxious frustration) that a single label cannot capture. Forcing a model to pick only one hard label discards useful information about ambiguity and intensity.

Moreover, human annotators often disagree when labeling datasets like MELD or IEMOCAP. These benchmarks treat this disagreement as noise and resolve it by majority voting, sometimes collapsing conflicting votes into a neutral category. But recent work argues that this disagreement reflects genuine perceptual ambiguity, which should be modeled rather than thrown away.

To address this, some researchers have started exploring soft labeling. For example, Hong et al. (2025) [7] tested whether large language models could output probability distributions over emotion classes instead of a single category. Using zero-shot and few-shot prompting on Gemini-1.5-Flash, they compared the distributions against human annotator opinions using KL divergence and Jensen-Shannon divergence. Their results suggest that soft labels can maintain good performance on traditional hard label benchmarks while capturing emotional nuance that hard labels miss entirely.

2.2 Monolithic Multimodal Fusion Versus Agentic Approaches

Traditional emotion recognition in conversations (ERC) combines text, audio, and video within a single neural network. These monolithic fusion models typically use one of three strategies:

1. **Late fusion (decision-level):** Train separate classifiers per modality and merge their predictions (e.g., by voting or averaging). This is simple but ignores cross-modal relationships.
2. **Token-level early fusion:** Project acoustic and visual features into a shared embedding space, concatenate them with text embeddings, and feed the result into a transformer. A common problem is modality collapse, in which the model learns to rely mostly on a dominant modality and ignores other modalities.
3. **Cross-attention / dense fusion:** Use cross-modal attention layers to align and fuse embeddings dynamically. The Multimodal Transformer (MulT) [8] is a well-known example.

Although monolithic fusion has been widely used, forcing one backbone to encode, align, and fuse all modalities in a single pass can overload it. This results in the model struggling to pick up subtle signal from different modalities.

2.2.1 Feature-to-Prompt Injection and the Semantic Gap

To avoid dense vector fusion, researchers tried pipeline approaches that turn non-text modalities into text features and append them to the LLM’s prompt. Early work used handcrafted acoustic features like eGeMAPS [9], which extracts 88 statistical parameters (pitch, jitter, formants, etc.) from raw audio. However, feeding raw numbers to an LLM creates a semantic gap: the model must first map physical measurements to emotional concepts. This works poorly, often worse than simply projecting raw audio into a continuous space.

A better solution came from using vision-language models to translate video frames into natural language descriptions. Zhang et al. (2025) [10] introduced DialogueLLM, which uses ERNIE Bot as a visual expert to describe video frames in plain text. The central LLM then receives these descriptions as part of the prompt, bypassing the need to interpret raw pixels or audio features. This turns multimodal fusion into a text-only reasoning task and outperforms dense vector fusion by about 5%.

Still, both eGeMAPS and DialogueLLM are static, single-pass pipelines. The LLM must handle a large prompt and resolve any conflicting signals in one forward

Model	Core Methodology	Key Highlight
<i>Pre-LLM Baselines</i>		
DialogueRNN COSMIC	Attentive Speaker GRUs Commonsense Graph Networks	Contextual recurrence Mental state tracking
<i>Track 1: Text-Only Generative LLMs</i>		
InstructERC LaERC-S CDEA	Retrieval-augmented LoRA Dynamic speaker cues Causality-driven commonsense	Prompt RAG Interlocutor traits Trigger extraction
<i>Track 2: Multimodal & Agentic LLMs</i>		
EmotionLLaMA DialogueLLM BeMERC AffectAgent	Instruction-tuned MLLM Frame textualization Video-derived behavior Multi-agent reinforcement	Emotion-specific encoders Visual semantic injection Micro-expression alignment Agentic conflict resolution

Table 2.1: Summary of representative ERC architectures.

pass, without any structured deliberation.

2.2.2 From Static Pipelines to Agentic Collaboration

Recent architectures try to overcome this by using multiple collaborative agents. **AffectAgent** [11] replaces the single reasoning step with three agents: a query planner, an evidence filter, and an emotion generator. Instead of forcing one network to fuse everything, these agents iteratively retrieve relevant examples, evaluate conflicting evidence, and debate before producing a final emotion label. The system is optimized with multi-agent reinforcement learning. This structured, multi-step approach avoids both the multitasking overload of monolithic models and the rigidity of single-pass prompt injection.

2.3 State-of-the-Art in ERC

Although multimodal and multi-agent systems are becoming more common, most benchmark results still come from text-only large language models. Transcripts carry the core semantic information of a conversation, so well-optimized text-only LLMs serve as strong baselines.

Before LLMs became popular, early ERC systems used deep sequential networks. Two notable examples are **DialogueRNN** [12], which uses Gated Recurrent Units (GRUs) to track speaker states, and **COSMIC** [13], which adds psychological commonsense knowledge to handle contextual ambiguity.

More recently, researchers have integrated these contextual and commonsense ideas into generative LLM architectures. As a result, current state-of-the-art results on datasets like MELD and IEMOCAP fall into two tracks: text-only LLMs (which focus on speaker and causal tracking) and multimodal or agentic architectures (which focus on cross-modal fusion).

2.3.1 Track 1: Text-Only LLMs

Lei et al. (2023) proposed **InstructERC** [14], which turns emotion recognition into a generative task. They use LLaMA-2-chat-7B with a retrieval module that finds similar dialogue turns using SBERT and adds them as few-shot examples. InstructERC also maps different emotion label sets into a unified set of nine classes using a method called Label Unification (UIME). To keep the model from drifting, they fine-tune with LoRA ($r = 16$) on auxiliary tasks like speaker identification and emotion impact prediction.

Fu et al. (2024) [15] pointed out that most ERC models ignore how speakers change over time. Their model **LaERC-S** uses LLM world knowledge to generate implicit speaker characteristics. It organizes these into nine dynamic cues, such as speaker intentions (`xIntent`) and reactions (`xReact`). The listener reaction (`oReact`) gives the largest performance gain. LaERC-S uses a two-stage fine-tuning process: first it trains the LLM to generate the speaker cues, then it trains it to output the final emotion label.

Zhang et al. (2025) [16] introduced **CDEA (Causality-Driven Dialogue Emotion Analysis)**. This model explicitly identifies emotional causes between past and current utterances. It separates self-induced from other-induced causes and maps them to ATOMIC commonsense variables. CDEA uses BERTScore to retrieve similar dialogue histories and a pretrained BART model to produce clean contextual explanations from messy dialogue. During LoRA fine-tuning, it adds an auxiliary task: the model must output both the emotion class and its dictionary definition from SentiWordNet.

2.3.2 Track 2: Multimodal and Agentic Architectures

Text-only models ignore audio and video cues. Adding those modalities requires solving the representation problems discussed in Section 2.

Some recent systems use Multimodal Large Language Models (MLLMs). **EmotionLLaMA** [17] is a basic example: it uses separate audio and video encoders, aligns the features in a shared space, and applies instruction tuning to improve multimodal reasoning. However, shared continuous spaces can still blur fine-grained physical cues.

To get around this, **BeMERC** [18] (Behavior-Aware MLLM) argues that raw visual embeddings are not enough. Instead, BeMERC extracts explicit behaviors from video, like facial micro-expressions, body language, posture, and converts them into text before feeding them into the MLLM. This reduces ambiguity in

cross-modal reasoning.

Finally, **AffectAgent** recognizes that even detailed behavior-aware prompts force a single model to resolve conflicting signals in one pass. AffectAgent instead uses multiple agents: a query planner, an evidence filter, and an emotion generator. These agents explicitly debate conflicting audio-visual evidence. Instead of just memorizing benchmarks, this multi-step reasoning shows better robustness on ambiguous or sarcastic dialogue turns. Multi-agent collaboration appears to be a promising direction for multimodal ERC.

This chapter has reviewed the key literature on emotion recognition, from psychological foundations to current ERC architectures. Three persistent gaps were identified: the loss of emotional ambiguity from hard labeling, the modality collapse of monolithic fusion, and the semantic gap of handcrafted acoustic features. The next chapter presents the proposed methodology to address these gaps.

CHAPTER 3. METHODOLOGY

Chapter 2 identified three limitations of existing ERC systems: hard labeling discards emotional ambiguity, monolithic fusion suffers from modality collapse, and handcrafted acoustic features create a semantic gap. This chapter presents the technical design of our proposed soft-label, multi-agent framework to address these limitations. It describes the system architecture, the three specialized component agents, the central resolver agent with its five-step reasoning framework, and the heuristic fusion baselines used for comparison.

3.1 System Architecture Overview

The core contribution of this thesis is a modular, soft-label-driven multi-agent system designed to handle emotional ambiguity and cross-modal conflict in conversations. Traditional monolithic multimodal models attempt to encode, align, and fuse disparate raw signals in a single pass of a dense Large Language Model (LLM). This approach imposes a significant computational and cognitive burden on the model by forcing it to simultaneously process raw features, reason across all modalities, and perform fusion in a single step, which frequently results in sub-optimal predictions.

Our proposed framework circumvents this limitation by formulating multimodal fusion as a text-based collaborative process. The architecture decomposes an input conversational turn into three specialized, unimodal component agents, whose outputs are subsequently synthesized by a central resolver agent. All agents utilize the Gemini-2.5-flash-lite model, leveraging its native end-to-end processing of text, audio, and video inputs. Figure 3.1 illustrates the overall data flow.

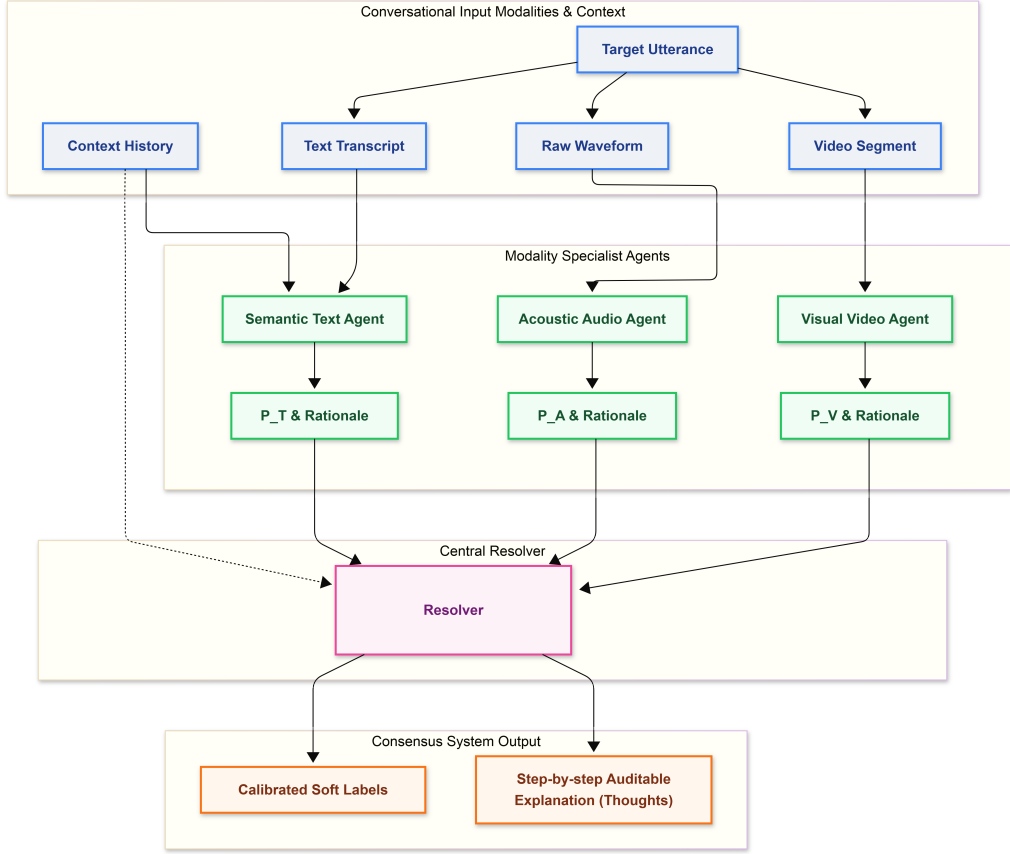


Figure 3.1: System architecture of the proposed soft-label multi-agent ERC framework.

Let a conversation be defined as $D = \{u_1, u_2, \dots, u_N\}$, where u_i represents the i -th conversational utterance. Each utterance u_i is accompanied by its raw acoustic waveform a_i , video frame sequence v_i , and textual transcript t_i . For a given target utterance u_i , we define its historical context of window size w as:

$$C_i = \{u_{i-w}, u_{i-w+1}, \dots, u_{i-1}\}. \quad (3.1)$$

The three component agents operate in parallel, each processing its respective modality independently. Instead of outputting discrete hard labels, each agent outputs two distinct components: a continuous probability distribution over the emotion classes (a soft-label distribution) that captures the model’s confidence and perceived ambiguity, and a concise natural language explanation (textual rationale) justifying its prediction based on modality-specific cues. These soft labels and rationales are subsequently transmitted to the central resolver agent, which performs structured, multi-step symbolic reasoning to produce the final emotional consensus.

3.2 Specialized Component Agents

3.2.1 The Semantic (Text) Agent

The Semantic Agent (T) models linguistic and contextual cues from the conversation transcript. It processes the target transcript t_i alongside the historical context window C_i to capture conversational flow, speaker relationships, and contextual shifts. Formally, the agent maps the textual inputs to a soft-label distribution P_T and a textual rationale R_T :

$$P_T, R_T = T(t_i, C_i), \quad (3.2)$$

where $P_T = [p_{T,1}, p_{T,2}, \dots, p_{T,K}]$ represents the probability distribution over the K emotion categories, satisfying $\sum_{j=1}^K p_{T,j} = 1$.

The prompt structure for the Semantic Agent enforces a rigorous output format: the model must analyze the conversational context, identify potential linguistic sarcasm or understatement, and output a strict JSON object containing the emotional probability distribution.

3.2.2 The Acoustic (Audio) Agent

The Acoustic Agent (A) analyzes the raw speech waveform a_i to evaluate vocal delivery. By directly processing the audio rather than relying on static, handcrafted feature sets like eGeMAPS, the agent successfully captures dynamic paralinguistic cues such as pitch variation, volume fluctuations, speaking rate, and non-verbal vocalizations (e.g., sighs or laughs). The agent operates through a structured, three-step reasoning process: it extracts the baseline acoustic features of the isolated clip, aligns these features with the literal transcript t_i to assess how delivery complements or contrasts with the text, and synthesizes these observations into an emotion probability distribution.

Crucially, the prompt forbids the agent from inferring broader conversational context, forcing it to remain a strict specialist in isolated vocal cues. It outputs a natural language rationale (R_A) and a soft-label distribution (P_A) formatted as a JSON dictionary summing exactly to 1.0. Formally, this is expressed as:

$$P_A, R_A = A(a_i, t_i), \quad (3.3)$$

where the transcript t_i is utilized solely for text-audio alignment rather than full semantic analysis.

3.2.3 The Visual (Video) Agent

The Visual Agent (V) analyzes the raw video segment v_i to capture the target speaker’s non-verbal behavior through a structured four-stage reasoning process.

To handle videos containing multiple individuals, the agent begins by locating and tracking the target speaker using a provided speaker identity. Once focused, it observes objective facial and bodily features, tracking key indicators such as eyebrow movement, eye shape, mouth positioning, and body posture. The agent then correlates these physical expressions with the six basic emotions and a neutral baseline, assigning an intensity score ranging from 1 to 5 based on characteristic cues-such as linking furrowed brows and a tight jaw to anger, or crinkled eyes and upturned lip corners to joy. Finally, these intensity values are synthesized into a balanced probability distribution over the complete emotion set.

The agent outputs both a textual rationale and a soft-label distribution formatted as a JSON dictionary that sums to 1.0. Formally, this operation is defined as:

$$P_V, R_V = V(v_i, s_i, t_i), \quad (3.4)$$

where P_V represents the soft-label distribution, R_V denotes the rationale, s_i is the speaker identifier, and t_i is the transcript provided strictly for contextual reference.

3.3 Resolver Agent Design

The central Resolver Agent (R) acts as a reasoning-driven meta-agent designed to synthesize the outputs of the three unimodal specialists. It processes the soft-label distributions and textual rationales generated by the text, acoustic, and visual agents, alongside the broader conversational history (C_i):

$$\mathcal{I}_i = ((P_T, R_T), (P_A, R_A), (P_V, R_V), C_i). \quad (3.5)$$

The primary objective of the Resolver is to compute a well-calibrated final consensus distribution, P_{Final} :

$$P_{\text{Final}} = R(\mathcal{I}_i), \quad (3.6)$$

where $P_{\text{Final}} = [p_{F,1}, p_{F,2}, \dots, p_{F,K}]$.

Rather than relying on traditional black-box vector projections or simple mathematical averaging, the Resolver operates entirely through symbolic textual reasoning. This design allows the resolver to explicitly analyze the textual rationales and soft-label vectors generated by each modality. By evaluating cross-modal consensus within this shared reasoning space, the Resolver can logically weigh conflicting

signals and identify the most globally coherent emotional probability distribution.

3.3.1 The Resolution Framework

The Resolver evaluates the multi-agent inputs through a structured four-step reasoning process:

1. **Cross-Modal Distribution Analysis:** The Resolver parses the soft-label vectors (P_T, P_A, P_V) alongside the textual rationales from each sub-agent to identify overlapping emotional tendencies across modalities, tracking candidate classes that exhibit broad cross-modal support.
2. **Weighted Confidence Integration:** The Resolver computes an aggregate confidence profile for each candidate emotion by summing its probabilities across all three modalities:

$$Score(e) = p_{T,e} + p_{A,e} + p_{V,e}. \quad (3.7)$$

During this step, decisive unimodal predictions (characterized by a sharp peak probability ≥ 0.5) are heavily prioritized, whereas highly diffuse or high-entropy distributions are treated as weaker signals.

3. **Prior-Driven Conflict Resolution:** In cases of explicit cross-modal contradiction, the Resolver applies domain-specific structural priors to break ties. The text agent is granted priority for semantically driven emotions (e.g., disgust derived from explicit insults), the acoustic agent takes precedence for intensity-based distinctions (e.g., distinguishing anger from sadness via acoustic energy), and the visual agent is prioritized for distinct physical expressions. The broader conversational context (C_i) serves as the final tiebreaker.
4. **Distribution Calibration:** Based on the final qualitative and quantitative consensus, the agent distributes the probability mass to reflect its confidence, concentrating weight on the chosen dominant emotions while maintaining a valid, calibrated probability vector across the remaining classes.

3.3.2 Output Format and Verification

To guarantee reliable downstream parsing, the Resolver is constrained to a strict JSON-encapsulated output block containing a step-by-step audit trail alongside the numerical vectors:

```
Thoughts: [Step-by-step reasoning covering agent
           agreement, conflicts, and final judgment]
Soft Labels: {
```

```

    "neutral": 0.05,
    "joy": 0.10,
    "sadness": 0.80,
    "anger": 0.05,
    "fear": 0.00,
    "surprise": 0.00,
    "disgust": 0.00
  }

```

This strict formatting ensures that the resulting distribution P_{Final} remains mathematically valid ($\sum p_{F,j} = 1.0$) while providing an explicit, human-auditable explanation for every conversational turn.

3.4 Heuristic Fusion Baselines

To benchmark our reasoning-driven resolver agent, we implement two classical late-fusion strategies that mathematically combine the outputs of the three component agents: mean fusion (for soft-label integration) and Borda count fusion (for hard-label voting).

3.4.1 Mean Fusion (Soft-Label Integration)

Mean fusion serves as the baseline mathematical approach for combining soft-label distributions. It assumes each modality carries equal weight, ignoring potential cross-modal inconsistencies or context. Let P_T, P_A, P_V be the K -dimensional soft-label distributions from the semantic, acoustic, and visual agents over the emotion set $\mathcal{E}$. The mean fusion distribution $P_{\text{Mean}} = [p_{\text{Mean},1}, \dots, p_{\text{Mean},K}]$ is computed as the element-wise arithmetic mean:

$$p_{\text{Mean},j} = \frac{1}{3} (p_{T,j} + p_{A,j} + p_{V,j}) \quad (3.8)$$

for each emotion class $j \in \{1, \dots, K\}$. Since each individual distribution (P_T, P_A, P_V) sums to one, the resulting P_{Mean} is also guaranteed to sum to one. This baseline preserves the soft-label format and carries forward the combined ambiguity of the sub-agents without forcing a premature hard choice.

3.4.2 Borda Count Fusion (Hard-Label Integration)

Borda count fusion is a consensus voting mechanism derived from social choice theory. Unlike mean fusion, which operates directly on raw probabilities, Borda count relies on the relative rank ordering of emotion categories predicted by each sub-agent. This strategy yields a single resolved hard label.

Let $\mathcal{E}$ be the set of K emotion categories. For each agent $a \in \{T, A, V\}$, the emo-

tions are sorted by descending probability to establish a rank $R_a(e) \in \{1, \dots, K\}$, where $R_a(e) = 1$ denotes the top-1 choice and $R_a(e) = K$ denotes the lowest. Points are assigned inversely as $K - R_a(e)$. The total Borda score for a given emotion e is formulated as:

$$Score_{\text{Borda}}(e) = \sum_{a \in \{T, A, V\}} (K - R_a(e)). \quad (3.9)$$

The final hard label $\hat{y}_{\text{Borda}}$ corresponds to the emotion that maximizes the total score:

$$\hat{y}_{\text{Borda}} = \arg \max_{e \in \mathcal{E}} Score_{\text{Borda}}(e). \quad (3.10)$$

In the event of a tie, the candidate class with the highest absolute probability assigned by the text agent is selected. While Borda count is highly robust to extreme outlier probabilities, it is incapable of outputting continuous soft-label distributions.

This chapter has described the proposed multi-agent architecture, including the three unimodal component agents (semantic, acoustic, visual) and the resolver agent with its structured reasoning framework, as well as the heuristic baselines used for comparison. The next chapter presents the experimental setup and numerical results.

CHAPTER 4. EXPERIMENT AND ANALYSIS

Chapter 3 presented the proposed soft-label, multi-agent framework. This chapter evaluates the framework empirically. It first describes the datasets, evaluation metrics, and experimental infrastructure. Then it reports baseline unimodal and heuristic fusion results, validates the soft-label hypothesis, compares the agentic resolver against monolithic architectures, and examines the impact of raw waveform audio processing. A qualitative analysis explains why and how the proposed method works.

4.1 Experimental Setup

4.1.1 Datasets

We evaluate the proposed framework on the test splits of two widely used Emotion Recognition in Conversation (ERC) benchmarks.

MELD (Multimodal EmotionLines Dataset) [1] is a multi-party dataset containing video clips, audio segments, and transcripts extracted from the television show *Friends*. MELD represents a challenging open-domain scenario with low signal-to-noise ratios, background laugh tracks, frequent speaker overlap, and sarcasm. The dataset is annotated with seven discrete emotion classes: neutral, surprise, fear, sadness, joy, disgust, and anger.

IEMOCAP (Interactive Emotional Dyadic Motion Capture) [2] is a dyadic conversation dataset consisting of long, structured interactions between pairs of actors. IEMOCAP requires deep contextual lookback to resolve emotional ambiguity, for example distinguishing frustration from anger or sadness. We evaluate using its six target classes: neutral, happiness, sadness, anger, frustration, and excitement.

4.1.2 Evaluation Metrics

We use standard discrete classification metrics together with information-theoretic divergence and calibration measures.

Accuracy (Acc) and **weighted F1-score** ($F_1(W)$) are used for hard-label classification tasks. Weighted F1 accounts for class imbalance by weighting each class’s F1 score by its support.

Top-2 accuracy (Acc@2) and **weighted F1@2** ($F_1(W)@2$) measure classification success when the ground truth label lies within the model’s top two highest-probability classes.

Jensen-Shannon Divergence (JSD) [19] measures the similarity between the model’s soft-label distribution Q and the ground truth distribution P (the statistical distribution of human annotator votes). JSD is a symmetric and bounded version of Kullback-Leibler divergence:

$$JSD(P \parallel Q) = \frac{1}{2}D_{KL}(P \parallel M) + \frac{1}{2}D_{KL}(Q \parallel M), \quad (4.1)$$

where $M = \frac{1}{2}(P + Q)$ is the mixture distribution. The KL divergence is defined as:

$$D_{KL}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log_2 \left(\frac{P(x)}{Q(x)} \right). \quad (4.2)$$

Using base-2 logs, JSD is bounded in $[0, 1]$, where 0 indicates identical distributions and 1 indicates completely disjoint distributions.

4.1.3 Model Backbones and Infrastructure

All specialized agents and the resolver are evaluated using Gemini-2.5-flash-lite. This model is selected for its native multimodal processing capabilities, allowing direct ingestion of raw audio waveforms and video keyframes alongside text tokens.

4.2 Baseline Performance

To establish empirical lower bounds, we evaluate independent unimodal agents and standard heuristic late-fusion strategies. Table 4.1a and 4.1b summarizes the results on the MELD test set for both hard-label and soft-label configurations.

Text remains the most predictive single modality, achieving 60.00% hard accuracy ($M - 01$). The visual modality ($M - 02$) and acoustic modality ($M - 03$) give baseline hard accuracies of 51.07% and 46.82%, respectively. Trimodal monolithic fusion ($M - 05$) achieves 59.04% hard accuracy, showing that merging all modalities inside a single model’s context window does not outperform the text-only baseline.

4.3 Soft-Label Validation (Hypothesis 1)

Our first hypothesis states that representing emotions as continuous probability distributions (soft labels) yields a systematic performance boost by preserving emotional ambiguity. As shown in Table 4.1, transitioning from hard classification to soft distribution outputs gives a consistent accuracy improvement across all single-modality baselines and the agentic framework.

For the text agent, accuracy increases from 60.00% to 62.41% (+2.41% absolute). For the visual agent, accuracy increases from 51.07% to 54.76% (+3.69%). For the acoustic agent, accuracy increases from 46.82% to 54.43% (+7.61%). For the agen-

Config ID	Modality	Architecture	Hard Acc	Hard F1
$M - 01$	T	Monolithic	60.00%	60.21%
$M - 02$	V	Monolithic	51.07%	49.51%
$M - 03$	A	Monolithic	46.82%	48.26%
$M - 05$	T+V+A	Monolithic	59.04%	59.57%
$M - 07$	T+V+A	Resolver Agent	52.41%	51.75%

(a) Hard-label classification performance.

Config ID	Modality	Architecture	Soft Acc	Soft F1	Acc @2	F1 @2	Avg JSD
$M - 01$	T	Monolithic	62.41%	62.54%	81.69%	81.37%	0.2440
$M - 02$	V	Monolithic	54.76%	52.17%	76.33%	74.69%	0.2867
$M - 03$	A	Monolithic	54.43%	55.39%	74.84%	74.98%	0.2857
$M - 05$	T+V+A	Monolithic	55.41%	56.17%	79.48%	79.46%	0.2792
$M - 06H$	T+V+A	Borda Count	63.51%	62.48%	82.31%	81.62%	-
$M - 07$	T+V+A	Resolver Agent	61.06%	60.81%	81.25%	80.77%	0.2492

(b) Soft-label and top-2 distribution evaluation performance.

Table 4.1: Modality and fusion baseline performance on the MELD test set partitioned by hard and soft evaluation metrics.

tic resolver, accuracy increases from 52.41% to 61.06% (+8.65%). This consistent improvement validates Hypothesis 1. Continuous distribution modeling preserves the uncertainty and mixed emotional states typical of human conversation, preventing models from committing to an incorrect hard choice when expressions are ambiguous.

Evaluating top-2 classifications reveals even stronger alignment. The soft-label text model achieves a top-2 accuracy of 81.69% ($F_1(W)@2 = 81.37\%$), the visual model reaches 76.33%, and the acoustic model reaches 74.84%. This high performance indicates that even when the top-1 prediction is wrong, the true emotion is often captured within the model’s second-highest probability mass.

4.4 Agentic vs. Monolithic Comparison (Hypothesis 2)

Our second hypothesis states that a modular multi-agent system coordinated by a reasoning-driven resolver outperforms monolithic, single-block multimodal models by using logical reasoning rather than mathematical vector fusion.

On MELD test set (Table 4.1), the soft-label agentic resolver ($M - 07$) achieves 61.06% accuracy and a weighted F1 of 60.81%. This outperforms the monolithic multimodal model ($M - 05$ at 55.41% accuracy and 56.17% F1) by absolute margins of 5.65% in accuracy and 4.64% in F1.

To test generalizability, we also evaluate the configurations on the IEMOCAP test set, which contains long dyadic interactions. Table 4.2 summarizes these re-

sults.

Config ID	Modality	Architecture	Accuracy	F1 (W)	Acc@2	Avg JSD
<i>I</i> – 01	T	Monolithic	50.43%	49.61%	76.08%	0.2387
<i>I</i> – 02	V	Monolithic	40.75%	36.14%	66.95%	0.2981
<i>I</i> – 03	A	Monolithic	41.99%	39.83%	64.98%	0.2900
<i>I</i> – 06 <i>M</i>	T+V+A	Mean Fusion	46.12%	43.64%	73.30%	0.2409
<i>I</i> – 06 <i>H</i>	T+V+A	Borda Count	46.30%	43.57%	71.39%	-
<i>I</i> – 07	T+V+A	Agentic Resolver	47.78%	45.61%	72.32%	0.3008

Table 4.2: Experimental results on the IEMOCAP test set.

On IEMOCAP, the agentic resolver (*I* – 07) achieves 47.78% accuracy and 45.61% F1, outperforming mean fusion (46.12%) and Borda count (46.30%). This demonstrates that symbolic reasoning-based resolution generalizes across different dataset structures.

The agentic resolver’s impact on probability distribution calibration reveals an important divergence between the two datasets. On MELD, the resolver achieves a Jensen-Shannon Divergence of 0.2492 relative to the human ground truth, outperforming the monolithic multimodal model (0.2792). This indicates that logical, text-level conflict resolution yields soft-label distributions that closely mirror human annotator disagreement patterns in short-form, open-domain conversations.

Conversely, on the IEMOCAP test set, the agentic resolver results in a higher JSD (0.3008) compared to the unimodal baselines and mean fusion (0.2409). This happens because the individual specialist models perform poorly on IEMOCAP’s long, complex dialogues, achieving only 40%-45% accuracy and F1-scores. Because these specialists provide highly noisy inputs, the resolver struggles to extract reliable information. This difficulty is clearly shown by the top-2 (@2) accuracy metrics. While MELD achieves 75%-81% @2 accuracy, IEMOCAP hovers around just 65% (with only the text modality reaching 75%).

4.5 Qualitative Analysis

While the quantitative results demonstrate that our hypotheses hold, this section addresses why they hold and how the system behaves in practice. We conduct three targeted qualitative investigations on the MELD and IEMOCAP test sets: a comparative analysis of hard versus soft label response patterns; a contrastive case study of monolithic versus agentic architectures; and a deep dive into the resolver agent’s symbolic reasoning chains to expose its conflict adjudication mechanics.

4.5.1 Why Soft Labels Outperform Hard Labels

Across all unimodal configurations on MELD, transitioning from hard-label to soft-label outputs consistently boosts top-1 accuracy. In the hard-label setting, instructing the model to commit to exactly one emotion class forces premature closure; it tends to commit to the easiest and often the first emotion that it recognizes, disregarding subtle clues and ambiguity. Conversely, soft-label prompting allows the model to distribute probability mass appropriately across all plausible classes, retaining ambiguity.

Example: Multi-Emotional Ambiguity (MELD Dialogue 4, Utterance 3). Consider the utterance "Oh my God!" spoken by Monica after Phoebe announces that her baby kicked. This exclamation naturally carries a mixture of two distinct emotions: astonishment (*surprise*) at the sudden physical sensation, and excitement (*joy*) for the positive milestone.

Under the hard-label configuration, the model is forced to commit to a single discrete class. It focuses on the positive long-term sentiment and predicts:

"Monica's utterance 'Oh my God!' is a strong exclamation reacting to this exciting news. The intent is to express astonishment and shared happiness at this significant milestone. Therefore, joy is the most appropriate emotion..."

Prediction: joy (incorrect; ground truth = surprise).

Under the soft-label configuration, the model is able to represent this emotional duality by distributing its probability mass:

"Monica's utterance, 'Oh my God!', is a strong exclamation reacting to this unexpected and positive news. The primary emotion is surprise at the event, closely followed by joy for Phoebe and the milestone."

Prediction: {surprise: 0.70, joy: 0.25, neutral: 0.01} (correct; ground truth = surprise).

The hard-label objective incentivizes finding the safest single label (often defaulting to neutral), whereas the soft-label objective forces the model to explain the distribution of plausible emotions by reasoning about speaker relationships and situational energy. Across the corpus, there are 267 utterances where the hard-label model is incorrect but the soft-label model is correct, compared to only 204 regressions. This net gain of +63 samples explains the +2.41% top-1 accuracy improvement for text and validates Hypothesis 1.

Furthermore, soft labels enable near-miss recovery. When the soft text model's

top-1 prediction is incorrect, the true emotion is frequently its second choice, as captured by the Acc@2 metric (achieving 81.69% top-2 vs. 62.41% top-1 accuracy). A wrong hard prediction provides zero downstream information, whereas a wrong soft prediction still calibrates uncertainty reliably for human-in-the-loop systems.

This behavioral divergence extends to other modalities. In the visual modality, hard models suffer from shallow feature-to-category mapping (e.g., misinterpreting the mouth shape of the spoken exclamation "Oh!" as a smile of joy). The soft visual model again manages to extract richer information about facial expression, body language, and even contextual information about other characters visible in the clip. Acoustically, hard models default to neutral when signals are subdued, treating the category as a garbage bin for low-intensity inputs. The soft acoustic model instead hedges its uncertainty, distributing probability mass to capture subtle paralinguistic details like hesitations and sighing markers.

4.5.2 Why the Monolithic Trimodal Model Fails

The monolithic trimodal model ($M - 05$) underperforms the text-only baseline ($M - 01$) on MELD in both hard (59.04% vs. 60.00%) and soft (55.41% vs. 62.41%) settings. Structural analysis reveals two root causes of monolithic failure.

Root Cause 1: Modality Blurring and Undifferentiated Fusion. Monolithic models process text, audio, and video simultaneously in a single forward pass. This high cognitive load diffuses the model’s attention, causing it to conflate cross-modal signals and latch onto surface-level visual cues, overriding critical acoustic or semantic details.

Consider the utterance "There you are!" spoken by Chandler in MELD Dialogue 124, Utterance 11 (ground truth: *surprise*). The monolithic model ($M - 05$) correctly notices the surprising nature of the encounter, but gets overwhelmed by the visual cue of Chandler running with a smile and open arms:

"...his final utterance, 'There you are!', combined with the visual evidence, strongly indicates relief and happiness upon finding her. The video shows Chandler running towards Janice with open arms and a wide smile... Therefore, the primary emotion is joy..."

Prediction: joy (incorrect; ground truth = surprise).

Conversely, the Agentic Resolver ($M - 07$) avoids this blurring by decoupling feature extraction from synthesis:

- **Specialist Agents** independently evaluate their assigned modalities: the Text Agent registers *joy* (0.70), but the Vision Agent detects *surprise* (0.60) based on the discovery, and the Acoustic Agent strongly detects *surprise* (0.71) based on Chandler’s sharp pitch rise and high vocal energy.
- **The Resolver** then synthesizes these inputs using symbolic rules, overriding the semantic text cue:

”...The AUDIO agent’s acoustic analysis (pitch rise, increased energy) strongly supports surprise. The VISION agent also supports surprise... While TEXT sees joy due to the playful context, the core event of discovery aligns better with surprise.”

This reasoning correctly resolves the cross-modal consensus to predict *surprise* (correct; soft-label prediction distribution: {surprise: 0.65, joy: 0.25, neutral: 0.05}).

Root Cause 2: Silent Modality Failures. A single corrupted or misaligned modality can derail the entire monolithic reasoning chain. This is particularly problematic in naturalistic datasets like MELD, where audio-visual segment boundaries can occasionally be misaligned or track the wrong speaker.

In MELD Dialogue 159, Utterance 1, Joanna is the annotated speaker of “Because if you feel uncomfortable with your friend dating someone you work for, there are always ways to fix...that” (ground truth: *anger*), but the corresponding video/audio clip in MELD’s raw data shows Ross speaking a different line in a resigned, neutral manner. The monolithic model ($M - 05$) is completely derailed by this cross-modal discrepancy:

”The prompt identifies ‘Joanna’ as the target speaker in the text, but the provided video and audio clearly show ‘Ross’. Assuming the analysis should be based on the multimodal data... tone is resigned and flat... predicting neutral.”

Prediction: neutral (incorrect; ground truth = *anger*).

In contrast, the Agentic Resolver ($M - 07$) delegates processing to independent specialists and uses symbolic reasoning to isolate the noise:

- **Specialist Agents** process their modalities independently. The Text Agent predicts *anger* (0.85) for Joanna’s textual threat, while the Vision (0.94 *neutral*) and Acoustic (0.83 *neutral*) agents evaluate Ross’s face and voice in the misaligned video clip.
- **The Resolver** analyzes the conflict and applies domain priors to isolate the

visual/acoustic failure:

"TEXT is most reliable for semantic emotions. The utterance 'there are always ways to fix...that' is a veiled threat, a semantic construct, strongly supporting TEXT's 'anger' interpretation... The neutral presentation observed by VISION and ACOUSTIC is likely a facade [or mismatch]... anger is the primary underlying emotion."

The resolver down-weights the noisy visual/acoustic inputs and outputs the correct emotion of *anger* (correct; soft-label prediction distribution: {anger: 0.65, neutral: 0.25, sadness: 0.06}).

4.5.3 Deep Dive: Resolver Agent Conflict Resolution Mechanics

On IEMOCAP, the independent semantic agent holds the highest predictive power (50.43%), followed by the acoustic (41.99%) and visual (40.75%) agents (Table 4.3). The resolver synthesizes these imperfect, conflicting signals through three primary symbolic adjudication patterns.

Agent Role	Accuracy (Independent)	JSD to Ground Truth
Semantic (<i>T</i>)	50.43%	0.2387
Acoustic (<i>A</i>)	41.99%	0.2900
Visual (<i>V</i>)	40.75%	0.2981

Table 4.3: Independent agent modality contributions on the IEMOCAP test set.

a, Pattern 1: Conflict Resolution via Modality-Specific Specialization

This pattern demonstrates the resolver’s capacity for conflict resolution by recognizing that distinct modalities excel at capturing specific categories of information. Structurally, the system treats the **text** modality as the primary authority for contextual and situational dynamics, the **audio** modality as the most reliable indicator of paralinguistic intensity and vocal strain, and the **vision** modality as the optimal source for physical expressions and micro-behaviors.

We can see this rule-based delegation in action during a polite corrective utterance:

- **Text & Vision Agents** predict *neutral* (0.85 and 0.90), capturing the literal semantic context and a calm facial posture.
- **Acoustic Agent** detects an underlying sigh and pitch fluctuations, predicting *frustration* (0.65).

Rather than allowing the majority consensus to completely suppress the acoustic signal, the resolver mediates the conflict by auditing the specific strengths of the inputs:

"TEXT is reliable for semantics [polite correction]... AUDIO is reliable for intensity: the 'sigh' ('ahh') and the overall tone strongly suggest mild frustration... the resolved distribution should reflect both."

The resulting output distributes probability across both *neutral* (0.67) and *frustration* (0.31), matching human annotations (JSD = 0.0071). By modeling these modality-specific strengths, the resolver preserves low-magnitude yet highly diagnostic acoustic cues without overriding the broader semantic baseline.

b, Pattern 2: Cross-Modal Consensus for Unambiguous Conflict

The female speaker responds: *"Okay, I'm sorry, but I just stood in this line for an hour. Can I - is there any way I can-"* Human annotators rated this {frustration : 1.00}.

All three agents converge on frustration as their top-1 class (T : 0.75, A : 0.75, V : 0.70), creating a strong cross-modal consensus. The disagreement lies in the secondary signal: the Acoustic Agent detects vocal exasperation and assigns residual probability to *anger* (0.20), while the Visual Agent observes a neutral facial expression consistent with the polite opening phrase and assigns residual probability to *neutral* (0.20). The Resolver reasons:

"Anger is supported by the ACOUSTIC agent (0.20), specialized for intensity, and is highly plausible given the context of wasted time. Neutral has support from the VISION agent (0.20), but in the context of a complaint about an hour-long wait, anger is a more accurate representation of the underlying intensity."

The Resolver outputs {frustration : 0.75, anger : 0.20}. While this distribution does not exactly match the one-hot human label {frustration : 1.00}, the residual probability is placed on *anger* rather than *neutral*. Given the conversational context (an hour-long wait), anger is a substantially more compatible secondary emotion with frustration than neutrality, which the Resolver correctly identifies as a surface artifact of the scripted polite opener. The resulting JSD of 0.0956 confirms that the distribution remains tightly concentrated on the annotated emotion, with the small deviation going toward a semantically congruent class. This demonstrates the context-informed pruning behavior of the Resolver: even when a modality provides a plausible secondary signal (visual neutrality), the Resolver uses conversa-

tional context to discriminate between genuine emotional content and performance artifacts.

c, Pattern 3: Visual Outlier Override on MELD

Beyond the IEMOCAP case studies, on the broader MELD corpus, the most frequent conflict pattern involves the Visual Agent producing a strongly isolated signal that is contradicted by both Text and Acoustic agents. Consider the utterance "Let's get that ball and really move, hey, hey, ho, ho" (GT = joy). The Vision Agent fires {anger : 0.70, neutral : 0.10} - likely because the speaker's intense, animated gesturing triggers the anger classifier - while both the Text Agent ({joy : 0.90}) and Acoustic Agent ({joy : 0.80}) strongly agree on joy. The Resolver's cross-modal consensus scoring yields:

- joy: $0.9 + 0.10 + 0.8 = 1.8$ (supported by 3 agents)
- anger: $0.002 + 0.70 + 0.03 = 0.732$ (1 agent, isolated)

The Resolver reasons: "*anger is strongly supported by VISION (0.70), but this signal is isolated (only 1 agent's top-2) and contradicted by TEXT and ACOUSTIC. The visual cues might represent intense focus rather than anger.*" The final output {joy : 0.80, surprise : 0.15} is correct (JSD = 0.075). This visual outlier override pattern illustrates that the Resolver does not treat modalities as equally reliable; it discounts isolated, high-confidence visual signals when two stronger modalities disagree - consistent with the structural visual reliability bias documented in Section 4.7.

4.6 Comparison against State-of-the-Art (SOTA) Models

To evaluate the empirical standing of our proposed framework, we benchmark our peak zero-shot configurations against prominent baseline models and recent state-of-the-art (SOTA) architectures. Tables 4.4 and 4.5 present the performance breakdowns on the IEMOCAP and MELD test sets, respectively.

Model / Configuration	Architecture / Modalities	Fine-Tuned?	F1 (Weighted)
CDEA	Text (Causality-Driven)	Yes	73.26%
LaERC-S	Text (Speaker Reaction)	Yes	72.40%
InstructERC	Text (Prompt-RAG)	Yes	71.39%
DialogueLLM	Multimodal (Video-to-Text)	Yes	69.93%
Ours	Text-only (Zero-shot)	No	49.61%
Ours	Agentic Resolver	No	45.61%

Table 4.4: IEMOCAP performance comparison against baselines and SOTA models.

Model / Configuration	Architecture / Modalities	Fine-Tuned?	F1 (Weighted)
DialogueLLM	Multimodal (Video-to-Text)	Yes	71.90%
CDEA	Text (Causality-Driven)	Yes	69.34%
LaERC-S	Text (Speaker Reaction)	Yes	69.27%
InstructERC	Text (Prompt-RAG)	Yes	69.15%
Ours	Text-only (Zero-shot)	No	62.54%
Ours	Agentic Resolver	No	61.06%

Table 4.5: MELD performance comparison against baselines and SOTA models.

4.7 Discussion

4.7.1 Key Findings and Empirical Anomalies

The experimental results validate our core framework: soft labels effectively map emotional ambiguity (yielding 2–8% gains on MELD), and textual multi-agent architectures successfully bypass the pitfalls of single-pass monolithic latent-fusion models.

However, a notable anomaly emerged on MELD: the monolithic model ($M - 05$) dropped significantly in performance when shifted to the soft-label setting (59.04% hard vs. 55.41% soft accuracy). As established previously, processing text, audio, and video simultaneously in a single forward pass overloads the monolithic model, leading to poor performance even in simple hard-label classification. Forcing this already overwhelmed architecture to perform the cognitively heavy task of generating a full, continuous probability distribution degrades the results even further. Output logs reveal that under these heavy multi-modal loads, the monolithic model spreads its probability mass too widely, producing high-entropy and poorly calibrated distributions. Conversely, the agentic resolver ($M - 07$) bypasses this limitation by isolating unimodal feature extraction from fusion; it protects the final distribution from entropy expansion, lifting its soft-label accuracy to 61.06% (a +8.65% absolute gain over its hard variant).

a, The Modality Drag and the Limits of Zero-Shot Multi-Agent Fusion

An apparent anomaly in Tables 4.2 and 4.1 is that our Agentic Resolver slightly underperforms the text-only zero-shot baseline on both corpora (45.61% vs. 49.61% on IEMOCAP, and 61.06% vs. 62.54% on MELD). Rather than a systemic failure, this exposes a fundamental challenge in zero-shot multi-agent fusion: a modality drag effect caused by error propagation.

We attribute this degradation to three factors. First, text dominates emotionally and most importantly, contextually, it provides the densest semantic signal,

while acoustic and visual specialists achieve only 40%-45% accuracy on IEMOCAP. When the Resolver weighs these weaker inputs symbolically, they act as noise that lowers the otherwise clean text baseline. Second, because the Resolver operates only on specialist outputs (soft labels and rationales) rather than raw features, any blurring or misclassification from the acoustic or visual agents propagates to the final decision—a "garbage-in, garbage-out" effect. Third, the Resolver lacks a learned fusion strategy; it relies on static, prompt-based priors rather than a dynamic gating mechanism that could adaptively detect and suppress unreliable modalities. Supervised models overcome this through fine-tuning, which learns to weight modalities based on their reliability or attention signal strength.

4.7.2 Discussion on Zero-Shot Framework vs. Supervised SOTAs

The comparative benchmarks presented in Section 4.6 outline a clear architectural divide in conversational emotion recognition: the trade-off between localized, supervised optimization and training-free, zero-shot symbolic reasoning. While our proposed framework underperforms the absolute performance peaks of recent state-of-the-art (SOTA) models, a structural analysis reveals critical insights into the limitations of zero-shot large language models (LLMs) and the benefits of our framework’s inherent explainability.

a, Mechanics of the Performance Deficit on IEMOCAP

As shown in Table 4.4, our peak configuration (*I-01*) experiences an absolute weighted F1 deficit of -23.65% compared to the causality-driven SOTA model, CDEA. This performance gap on the IEMOCAP corpus is driven by two key algorithmic constraints.

First, there is a supervised adaptation premium. Top-tier models such as CDEA, LaERC-S, and InstructERC leverage supervised fine-tuning (SFT) or parameter-efficient Low-Rank Adaptation (LoRA). This fine-tuning explicitly teaches the network to model dataset-specific structural behaviors and actor-specific baseline tendencies unique to IEMOCAP’s dyadic setup. Our frozen backbone (Gemini-2.5-flash-lite), operating completely zero-shot, must rely exclusively on contextual in-context hints, making it inherently harder to maintain accurate emotional tracking across long conversational paths.

A critical limitation of our training-free framework is its inability to model fine-grained emotional granularity and rapid temporal dynamics. Because the frozen backbone lacks target-domain parameter tuning, it systematically confuses similar-intensity emotions (e.g., happiness vs. excitement, frustration vs. anger vs. sadness) on IEMOCAP, capturing broad arousal signatures rather than specific behav-

ioral nuances. This coarse representation also produces emotional inertia, where the system struggles to track rapid valence shifts over extended dialogue turns, failing to capture micro-fluctuations without the sequential priors that supervised fine-tuning provides.

b, The Narrowing Divide on Short-Form MELD Transcripts

In contrast to the wide margins observed on IEMOCAP, the gap between our methodology and supervised SOTAs shrinks substantially on the multi-party MELD benchmark (Table 4.5). Our soft-label monolithic configuration (*M-01 Soft*) narrows the performance gap to single digits, sitting at just -9.36% below the multi-modal SOTA, DialogueLLM, and within -6.70% of dominant fine-tuned text-only frameworks.

This closing divide highlights a core finding: short-form, open-domain utterances (such as the sitcom interactions in MELD) depend much more heavily on immediate semantic understanding and localized context than on long-range history tracking. Moreover, MELD’s six emotion classes are relatively distinct and well-separated compared to IEMOCAP’s more subtle distinctions (e.g., excitement vs happiness), reducing the likelihood of cross-modal confusion. The dataset’s instantaneous, reaction-based dialogue structure-typical of television comedy-also avoids the prolonged, gradually shifting emotional trajectories found in IEMOCAP, meaning our model does not suffer from the same emotional inertia or tracking delays. By shifting from a rigid hard-label choice to a continuous soft-label distribution (*M-01 Soft*), our system captures human annotator ambiguity out-of-the-box, reclaiming $+2.33\%$ in F1 margin over our own hard baseline without updating a single internal parameter. This proves that continuous probability distribution projections can mimic a portion of the performance gains traditionally unlocked only through expensive supervised training.

c, The Paradigm Trade-off: Performance versus Explainability

Ultimately, evaluating our zero-shot multi-agent system against supervised SOTAs forces a broader architectural choice between raw benchmark maximization and systemic transparency.

SOTA networks like DialogueLLM fuse multimodal elements via dense latent projections. While mathematically optimal for squeezing out peak accuracy on specific training splits, they produce opaque decisions that cannot be easily verified by humans, unless specifically trained to explain their choice.

Our multi-agent framework explicitly trades a margin of raw benchmark performance for architectural interpretability. By isolating modalities into independent

specialist agents and forcing a text-based, symbolic resolution step, our system generates a step-by-step, human-readable audit trail (the `Thoughts` field). This design ensures that every final probability distribution is backed by explicit, logical deductions, providing a level of error isolation and trustworthiness that monolithic SOTA black-boxes cannot achieve.

This chapter has presented the experimental results, including quantitative benchmarks and qualitative case studies, confirming the benefits of soft labeling, agentic reasoning, and raw waveform processing. The next chapter concludes the thesis and discusses future work.

CHAPTER 5. CONCLUSIONS

This chapter concludes the thesis by summarizing the core research problems, the proposed solutions, and the main findings from the experimental evaluation. It also discusses the limitations of the current work and outlines directions for future research.

5.1 Summary of Work

This thesis addressed the problem of multimodal emotion recognition in conversations (ERC), focusing on two fundamental limitations of existing approaches: the loss of emotional ambiguity caused by hard-label classification, and the cognitive overload and opacity of monolithic multimodal fusion architectures.

To address these limitations, we proposed a soft-label, multi-agent framework that decomposes the ERC task into three specialized unimodal agents (text, acoustic, visual) coordinated by a central resolver agent. Unlike traditional systems that fuse raw signals in a single dense representation space, our framework treats multimodal fusion as a symbolic, text-based reasoning problem. Each component agent outputs a probability distribution over emotion classes (soft labels) together with a natural language rationale. The resolver then synthesizes these inputs through structured deliberation, explicitly weighing cross-modal consensus and resolving conflicts using modality-specific priors.

We evaluated the proposed framework on the MELD and IEMOCAP benchmarks using the Gemini-2.5-flash-lite model. The experimental results confirmed the two core hypotheses of this thesis.

5.2 Summary of Contributions

This thesis makes three primary contributions to the field of affective computing.

First, empirical validation of the soft-label paradigm. We demonstrated that converting hard labels into continuous probability distributions yields systematic improvements in classification accuracy across all unimodal and multimodal configurations. On MELD, soft labeling improved text agent accuracy by +2.41%, visual agent accuracy by +3.69%, and acoustic agent accuracy by +7.61%. The agentic resolver gained +8.65% when shifting from hard to soft outputs. These gains are explained by the soft-label model’s ability to preserve ambiguity and enable near-miss recovery, as shown in the qualitative analysis.

Second, development of a reasoning-based resolver agent. We proposed and

validated an agentic coordination framework that replaces black-box vector fusion with symbolic, text-based reasoning. On MELD, the agentic resolver achieved 61.06% soft accuracy, outperforming monolithic fusion (55.41%) by an absolute margin of 5.65%. On IEMOCAP, the resolver achieved 47.78% accuracy, outperforming mean fusion (46.12%) and Borda count (46.30%). The qualitative analysis revealed that the resolver’s advantage stems from two structural properties: modality blurring is avoided by independent specialist agents, and single-modality failures are contained rather than corrupting the entire prediction.

5.2.1 Limitations

Several limitations of the current work should be acknowledged.

a, API Cost and Inference Latency

The proposed framework relies on cloud-based API calls to Gemini-2.5-flash-lite for each agent and each individual utterance. This introduces substantial monetary costs and significant inference latency, making the current system impractical for real-time deployment or high-throughput applications. Since evaluating a single conversational turn requires four separate API queries (three parallel unimodal specialist streams and one central resolver pass), the architecture scales poorly when applied to larger operational pipelines.

b, The Zero-Shot Performance Ceiling

A major limitation identified is the absolute performance ceiling faced by our training-free, unsupervised framework when compared against fine-tuned state-of-the-art (SOTA) systems. Operating completely zero-shot without localized weight adaptation introduces a performance deficit—most notably on the long-form, dyadic interactions of the IEMOCAP benchmark, where our framework falls −23.65% below the peak supervised SOTA model. Because our system uses a frozen backbone, it lacks the specialized, domain-tuned sequential priors that allow supervised architectures (e.g., CDEA, LaERC-S) to efficiently compress long dialogue histories and model target-actor characteristics. This suggests that pure zero-shot multi-agent prompting struggles to handle highly complex, extended conversational arcs without localized parameter-efficient fine-tuning.

c, Monolithic Soft-Label Degeneration

An unexpected anomaly emerged during the empirical evaluation: the monolithic multimodal model experienced a substantial performance drop when transitioned to a soft-label setting, falling from 59.04% hard accuracy to 55.41% soft accuracy on MELD. While we hypothesize that this stems from severe entropy expansion and overly diffuse distribution spreads under high-dimensional multi-

modal noise, this structural drop demonstrates that the soft-label paradigm is not a universal performance accelerator. Its mathematical viability depends entirely on architectural decomposition, meaning that forcing a single dense network block to simultaneously extract features and output complex continuous distributions degrades performance.

d, Dataset and Domain Bias

The empirical scope of this thesis is constrained by the specific characteristics of its evaluation corpora. MELD is derived from a single situational comedy (*Friends*), introducing biases such as background laugh tracks, recurring actor-specific mannerisms, and a heavily stylized, exaggerated comedic pacing. Conversely, IEMOCAP consists of scripted and acted studio interactions rather than spontaneous, organic everyday conversations. Consequently, how effectively this zero-shot multi-agent framework generalizes to completely naturalistic, unconstrained, or in-the-wild conversational environments remains unverified.

e, Generalization to Alternative Large Language Models

Finally, all specialist and meta-reasoning agents within this framework were evaluated exclusively with the Gemini-2.5-flash-lite backbone, selected for its native end-to-end processing of audio waveforms and video frames. It remains unclear whether the structural behaviors observed in this thesis—such as the visual outlier override pattern or text-based resolution rules—would hold true across other large language models. This is particularly relevant for smaller, locally hosted open-source models or architectures that require external, static token injection pipelines to ingest non-text modalities.

5.3 Future Work

Several directions for future research emerge from this work.

Knowledge distillation for local deployment. The most pressing practical limitation is API cost and latency. Future work should investigate distilling the agentic reasoning traces into a compact, locally deployable model. The resolver’s audit trails (the `Thoughts` field) provide a rich source of supervised data for training a smaller model to mimic the multi-agent deliberation process. A distilled model could operate without cloud dependencies, enabling real-time applications.

Irony and sarcasm modeling. The poor performance on sarcastic utterances points to a specific need for irony-aware prompting. Future iterations could incorporate few-shot examples that explicitly distinguish playful sarcasm from hostile sarcasm, or could add a dedicated sarcasm detection agent that flags potential in-

version before the resolver applies its standard priors.

Adaptive modality weighting. The current resolver uses fixed priors (text priority for semantics, audio for intensity, vision for physical expressions). A more sophisticated version could learn adaptive weights from data, or could condition its weighting on the detected conversational context (e.g., down-weighting vision in low-light conditions).

Robustness to missing modalities. The current resolver assumes all three modalities are available. In real-world settings, audio or video may be missing (e.g., text-only chat logs, audio-only phone calls). Future work should extend the framework to handle arbitrary modality subsets gracefully, perhaps by training the resolver to operate on variable-length agent inputs.

5.4 Closing Remarks

This thesis presented a soft-label, multi-agent framework for multimodal emotion recognition in conversations. The core idea is simple: instead of forcing a single model to fuse everything at once, decompose the task into specialist agents and let a resolver reason about their outputs in text. The empirical results on MELD and IEMOCAP confirm that this approach outperforms monolithic fusion, preserves emotional ambiguity through soft labels, and produces interpretable decisions via explicit reasoning chains. While limitations remain – particularly around cost, sarcasm, and generalization – the framework establishes a promising direction for building more transparent and robust affective computing systems.

This chapter has summarized the thesis, restated its contributions, discussed limitations, and outlined directions for future work.

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